Hao Wu | Resume

Data Scientist/Software Engineer

Status: Freelance Data Scientist/AI Engineer
Field: Software Engineering, Data Science
Techs: Python, R, JavaScript, HTML, CSS

Didcot, UK

Summary

Data Scientist & Software Engineer with extensive experience in full-stack application development. Expert in operationalising machine learning models and building interactive, data-driven applications using Python and R. Specialized in Generative AI, including the design of RAG pipelines, multimodal embedding strategies, and high-performance vector search optimization. Adept at managing the full lifecycle of data products from research and prototyping to deployment and monitoring.

Professional Experience

Data Scientist/AI Engineer - Freelance — Dec 2025 - Present

* Built an AI-powered outfit recommendation system using LangGraph for agent orchestration, implementing conversational workflows with tool-calling capabilities, Postgres checkpointing for session persistence, and Langfuse integration for observability and cost monitoring.
* Designed and deployed a vector search pipeline using Qdrant and FashionSigLIP embeddings, enabling semantic similarity search across wardrobe items with natural language queries.
* Developed production-ready microservices with FastAPI and Docker, including image preprocessing pipelines (background removal), metadata extraction via Gemini 2.5 with structured outputs, and RESTful endpoints for wardrobe management
* Implemented end-to-end document embedding workflows: raw image ingestion -> preprocessing -> VLM metadata extraction -> fashion-specific embedding generation -> vector database indexing, with CLI scripts for batch processing
* Built a deep learning model with transformer architecture as a Context-Aware Compatibility Re-ranker for Fashion Outfits

Data Scientist - Argus Media — Jan 2022 - Dec 2025

* Co-developed "Ask Argus" a production RAG system: Led the critical migration of the vector database layer to OpenSearch to improve scalability and retrieval latency.
* Successfully built and delivered a web-based solution for optimising fuel mixing worth č250,000
* Building production ready Shiny dashboards for commodity traders using {golem}framework
* Building custom Shiny inputs using JavaScript to enhance user experience
* Manage user permission and data storage with MySQL and AWS S3
* Contributing to the data preparation pipeline that feeds into the models forecasting commodity prices and distribution

Senior Data Science Specialist - Ricardo Energy & Environment — Jan 2020 - Jan 2022

* Building R Shiny applications integrated with SQL database for data checking and ratification
* Building bespoke R packages to simplify the process of data analysis
* Using regression and tree based statistical models to predict air quality trends
* Building Python CLI tool to scrape and compare data from XML files

Data Science Specialist - Ricardo Energy & Environment — Apr 2017 - Jan 2020

* Mentoring staff on doing data analysis in R
* Dynamic reporting, data visualisation and building tools for analysing air quality data

Education

Ph.D. - University of Edinburgh, UK — 2017

* Understanding the spatial-temporal variability of urban air pollution
* Dispersion and statistical modelling of urban air pollution
* Conducting air quality measurements using stationary and portable air quality monitors.

BSc (1st Class) - University of Edinburgh, UK 2013

* Environmental and Sustainable Chemistry

BSc - South China University of Technology, China 2013

* Applied Chemistry

Personal Projects

Agent for writing Environment Impact Assessment report - Github 2025

* Multi-agent AI system for automated EIA report drafting: LangGraph-based workflow coordinating Planner, Interviewer, and Drafter agents to generate Chinese Environmental Impact Assessment reports section-by-section
* Interactive information gathering and validation: Interviewer agent collects project details through conversation, validates completeness against regulatory guidelines, and signals readiness to draft

CLI Scanner for stocks with upcomming earnings events - Github 2025

* Scrape Market Chemeleon to get historical earnings performance
* Using ib_async to fetch live option data to assess the liquidity of the stock options

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